Science

Class 7 (English Medium)

Solved Exercises — Sindh Textbook Board

12 lessons — 192 questions

Human Organ Systems

1. (i) Which of the following juices comes out of the liver?

(a) Pak

(b) Gastric juice

✓ (c) Bile

(d) Pancreatic juice

1. (ii) In which organ does the chemical digestion of protein begin?

(a) In the mouth

(b) In the oesophagus

✓ (c) In the stomach

(d) In the small intestine

1. (iii) Which is the correct order of the organs of the digestive tract?

(a) Mouth ← Oesophagus ← Small intestine ← Stomach

(b) Pharynx ← Stomach ← Pancreas ← Liver

(c) Pharynx ← Liver ← Pancreas ← Small intestine

✓ (d) Mouth ← Oesophagus ← Stomach ← Small intestine

1. (iv) Which organ stops food from entering the windpipe while eating?

(a) Pharynx

(b) Oesophagus

✓ (c) Epiglottis

(d) Tongue

1. (v) In which row is the correct difference between breathing and respiration given?

(a) Breathing needs oxygen, respiration does not need oxygen.

(b) Carbon dioxide is breathed out in breathing, but not in respiration.

(c) In breathing air goes in, in respiration air goes out.

✓ (d) In breathing energy is used up, in respiration energy is released.

Question: 2. (a) Why are the liver and the pancreas not counted as part of the digestive tract?

Answer: Because food does not pass through them. They are accessory organs which only send juices into the digestive tract, so they are not counted as part of the digestive tract.

Question: 2. (b) Why are fatty foods not digested in the mouth or the stomach?

Answer: Because neither the saliva nor the gastric juice breaks down fat. Fat is first broken into small droplets by bile and then digested in the small intestine by the enzymes of the pancreatic juice.

Question: 2. (c) Why does the absorption of the parts of food take place only in the small intestine?

Answer: Because the walls of the small intestine have tiny projections (villi) and blood capillaries through which the digested food is absorbed into the blood. No other organ has this structure.

Question: 2. (d) Why is the act of breathing a physical process?

Answer: Because in it air only moves in and out, no chemical change takes place, and no enzyme takes part.

Question: 2. (e) Why does polluted air cause diseases of breathing?

Answer: Because the dust, smoke and germs present in polluted air damage the respiratory system, which causes diseases like cold, asthma and cough.

Question: 2. (f) Why should one not talk while eating?

Answer: Because while talking, the passage of the windpipe stays open, and a piece of food may go into the windpipe, which risks coughing or choking. During swallowing, the epiglottis closes the windpipe.

Question: 3. (i) What is physical or mechanical digestion? Describe the

process of physical digestion in the digestive tract.

Answer: Physical or mechanical digestion is the breaking of food into small pieces without any chemical change. In the mouth the teeth chew the food into smaller pieces, and in the stomach the muscular walls make mixing waves that mix the food with gastric juice and soften it.

Question: 3. (ii) Describe the chemical digestion of a fried potato (starch and fat) as it passes through the following organs, and also write the enzyme word equations: (a) Mouth (b) Stomach (c) Small intestine

Answer: (a) Mouth: the salivary enzyme starts changing starch into glucose. Word equation: Starch + salivary enzyme → Glucose. (b) Stomach: no chemical digestion of starch or fat takes place; the gastric juice only kills germs and starts protein digestion. (c) Small intestine: bile breaks fat into small droplets, and the enzymes of the pancreatic juice change starch into sugar and fat into fatty acids and glycerol. Word equations: Starch + pancreatic enzyme → Sugar; Fat + bile + pancreatic enzyme → Fatty acids + Glycerol.

Question: 3. (iii) What steps should be taken to avoid the diseases of the respiratory system?

Answer: Eat fresh fruits and vegetables, exercise regularly, live in a clean environment, avoid pollution, get proper vaccination, and avoid smoking.

Question: 3. (iv) What is the process of gaseous exchange?

Answer: In gaseous exchange, oxygen from the air in the tiny sacs (alveoli) of the lungs is absorbed into the blood capillaries, and carbon dioxide from the blood is absorbed into the air of the sacs, which is then breathed out.

Question: 3. (v) Describe the main functions of enzymes.

Answer: Enzymes speed up chemical processes and break large molecules into smaller molecules so that food can be digested. For example, the salivary enzyme changes starch into glucose.

Question: 4. (a) Prepare a diagram of the digestive system and mark the correct positions of the pancreas and the liver in it.

Answer: (Note: this diagram is described in words.) Draw the diagram like this: at the top the mouth, below it the oesophagus (long tube), below it the stomach (J-shaped sac). To the upper right of the stomach the liver (large reddish organ) and below it the gallbladder (small green sac). Below and behind the stomach the pancreas (long, thin organ). After the stomach the small intestine (coiled tube) in the middle, around it the large intestine, below it the rectum and finally the anus. Labels: mouth, oesophagus, stomach, liver, gallbladder, pancreas, small intestine, large intestine, rectum, anus.

Question: 4. (b) Draw a diagram of the process of gaseous exchange inside the human body.

Answer: (Note: this diagram is described in words.) In the diagram show one tiny sac (alveolus) of the lungs with a blood capillary around it. Show with arrows: oxygen goes from the air of the sac into the blood of the capillary, and carbon dioxide comes from the blood of the capillary into the air of the sac. Labels: alveolus (air-filled sac), blood capillary, oxygenated blood, deoxygenated blood, oxygen arrow (sac → blood), carbon dioxide arrow (blood → sac).

Question: Project (i) Explain your observation.

Answer: In the airtight bottle the living cockroach stays alive and shows signs of breathing, while in the bottle with the dead cockroach no such activity is seen. This proves that animals breathe.

Question: Project (ii) Why did we use a specimen animal (cockroach) in this experiment?

Answer: Because the cockroach is a small animal which fits easily into the conical flask, and its breathing can easily be observed.

Question: Project (iii) Why did we use airtight conical flasks?

Answer: So that outside air cannot enter, and any change in the air inside the bottle is only because of the animal's breathing.

Transport in Humans and Plants

Question: 1. 1 — Match "also produces blood pressure" with the correct statement in column "B".

Answer: 1 — (d) High blood pressure (Hypertension).

Question: 1. 2 — Match "absorb water from the soil" with the correct statement in column "B".

Answer: 2 — (c) Root hairs.

Question: 1. 3 — Match "is caused by blockage in the arteries" with the correct statement in column "B".

Answer: 3 — (e) Heart attack.

Question: 1. 4 — Match "carries blood to the whole body" with the correct statement in column "B".

Answer: 4 — (b) Aorta.

Question: 1. 5 — Match "carries water in plants" with the correct statement in column "B".

Answer: 5 — (a) Xylem.

2. (a) Deoxygenated blood from the body reaches the _____ of the heart.

Answer: right atrium (right upper chamber)

2. (b) Deoxygenated blood goes from the heart to the lungs through the _____.

Answer: pulmonary artery.

2. (c) The force that pulls water from the roots to the leaf cells is called _____.

Answer: the transpiration pull.

2. (d) Replacing a damaged organ of the body with a healthy organ is

called _____.

Answer: transplantation.

2. (e) The long tubes that carry water to the roots, stem and leaves are called _____.

Answer: xylem vessels.

Question: 3. (a) Why are there valves in the veins?

Answer: Because the blood pressure in veins is low and the blood has to return to the heart; the valves stop the blood from flowing backwards and let it flow only towards the heart.

Question: 3. (b) Why is there oxygen-rich blood in all arteries except the pulmonary artery?

Answer: Because all arteries carry oxygen-rich blood from the heart to the body, except the pulmonary artery, which carries deoxygenated blood from the heart to the lungs.

Question: 3. (c) Why does a heart attack happen when the arteries narrow?

Answer: Because cholesterol and fatty substances collect in the walls of the arteries and narrow them, which stops the blood supply to the heart muscles, and the part of the heart starts dying — that is a heart attack.

Question: 3. (d) Why does the flow of sap in the xylem have no particular direction?

Answer: According to the book, water rises through the dead, hollow and continuous xylem vessels from the roots to the stem and leaves because of root pressure and the transpiration pull — that is, the flow runs in one direction, from the roots to the leaves. (The book gives no explanation for a "no particular direction" flow.)

Question: 3. (e) Why are the walls of the lower chambers (ventricles) of the heart thicker than the walls of the upper chambers (atria)?

Answer: Because the ventricles have to push the blood out with force — the right ventricle pumps blood to the lungs and the left ventricle pumps blood to the whole body — while the atria only receive blood, so their walls are thin.

Question: 4. Prepare a labelled diagram of the heart and show the flow of blood in it clearly.

Answer: (Note: this diagram is described in words.) In the diagram show the four chambers of the heart: at the top the right atrium and left atrium, below them the right ventricle and left ventricle. Labels: vena cava, pulmonary artery, pulmonary vein, aorta, tricuspid valve, bicuspid (mitral) valve, pulmonary valve, aortic valve. Flow with arrows: Body → vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta → body.

Question: 5. (a) State the difference between arteries and veins.

Answer: Arteries carry blood from the heart to the body, their walls are thick and muscular, the blood pressure in them is high, and they have no valves; veins bring blood back from the body to the heart, their walls are thin, the blood pressure in them is low, and they have valves. (Exceptions: the pulmonary artery carries deoxygenated blood and the pulmonary veins carry oxygenated blood.)

Question: 5. (b) State the difference between atria and ventricles.

Answer: The atria are the two upper chambers of the heart which receive blood, and their walls are thin; the ventricles are the two lower chambers which push blood out with force, and their walls are thick.

Question: 5. (c) State the difference between oxygenated blood and deoxygenated blood.

Answer: Oxygen-rich blood has more oxygen and less carbon dioxide, and it goes from the lungs to the heart and the body; deoxygenated blood has

less oxygen and more carbon dioxide, and it goes from the body to the heart and the lungs.

Question: Project 1. What kind of tissues of the plant stem were cut and removed?

Answer: The ring of outer bark together with the phloem tissues of the stem was cut away, while the inner wood (xylem) was left intact.

Question: Project 2. What did you see at the cut part of the plant?

Answer: The part above the cut ring swelled up (bulging), because the food coming from the leaves collected there; in the other plant (which was not cut) no such change was seen.

Question: Project 3. What information about phloem was obtained from this experiment?

Answer: The experiment showed that the phloem carries out the transport of food inside the plant — food reaches from the leaves to the other parts of the plant through the phloem.

Question: Project 4. What effect did the cut have on the lower part?

Answer: Because the cut blocked the phloem path, food from the leaves could not reach the lower part, and that part became weak and wilted.

Question: Project 5. What conclusion did you reach?

Answer: Conclusion: the phloem transports food in the plant; when the phloem path is cut, food does not reach below and collects above the cut.

Reproduction and Growth in Plants

Environment and Its Effects on Living Things

Question: 1. (i) What is a habitat? Name the different kinds of habitats and describe their special features.

Answer: A habitat is the environmental region in which an organism lives and where it gets food, shelter, protection and facilities to reproduce. It has two main kinds: (1) Aquatic habitat — fresh-water (rivers, streams, ponds), sea-water (sea, salty water) and estuarine (where the river meets the sea); (2) Terrestrial habitat — forest (heavy rain, dense trees), grassland (grassy plains), desert (very little rain, hot and dry) and tundra (covered with snow, very cold). Every habitat has its own special conditions of temperature, light, moisture, soil and water.

Question: 1. (ii) How do organisms adapt to the changing conditions of the environment?

Answer: Organisms adapt to the environment with the help of their special characteristics (adaptation), such as changing colour or appearance (camouflage), moving from one place to another (migration), long winter sleep (hibernation), long summer sleep (aestivation), body covering, and leaf-shedding in plants. For example, the earthworm lives in burrows in the ground and is adapted to its environment.

Question: 1. (iii) Explain the following terms: (a) Ecosystem (b) Population (c) Community

Answer: (a) Ecosystem: the combined system of the living (biotic) part and the non-living (abiotic) part of a place, in which both interact with each other. (b) Population: a group of organisms of the same species living in a particular habitat. (c) Community: all the different populations (plants, animals and micro-organisms) living together in one habitat.

Question: 1. (iv) Which factors cause the daily and yearly changes of a habitat?

Answer: Daily changes: the difference between day and night, and the difference in light intensity and temperature — more light and heat in the day, less at night — because of which some animals are active in the day (diurnal) and some at night (nocturnal). Yearly changes: the coming of the four seasons, differences in temperature and rainfall through the year, and natural disasters (drought, flood, earthquake) and human activity, such as global warming.

Question: 1. (v) Describe the characteristics of some water animals which make them suited to living in a water habitat.

Answer: The organs of animals living in water are made according to the water environment: they obtain oxygen and food from the water itself, their bodies are made to swim and move in water, and some animals, like whales and sharks, live only in water.

Question: 1. (a) Describe how animals and plants adapt to different conditions in order to live in a place.

Answer: The conditions of every place are different — light, temperature, moisture, air, soil and water — and animals and plants take on special features to adapt to those conditions: animals through camouflage, migration, hibernation, aestivation and body covering, and plants through shedding leaves and changing their structure. For example, the skin colour of people living in hot regions helps protect them from the heat of the sun, and in cold regions animals have thick fur or a fatty covering.

Question: 1. (b) What are a food chain and a food web? Give two examples of each.

Answer: Food chain: a series of organisms in which each organism gets food and energy from the previous organism. Examples: (1) Grass →

Grasshopper → Frog → Snake → Hawk; (2) Grass → Rat → Hawk. Food web: a web made of many food chains joined together, because one organism can get food from several sources. Examples: (1) Grass and plants → Grasshopper → Frog → Hawk, and grass and plants → Rat → Hawk; (2) Grass and plants → Crow → Snake → Hawk, and grass and plants → Crow → Hawk.

Question: 1. (c) Why do people change their place of living (habitat)? What should be done to protect any habitat?

Answer: People bring changes in habitats for their needs — for example, using land for settlements and farming — and their activities create pollution and global warming, which change habitats. To protect a habitat, pollution must be reduced, forests and natural areas must be saved from destruction, and the balance of the environment must be kept. (The book gives no detailed conservation list; this answer is drawn from the book's statements on human activity, pollution and global warming.)

2. (i) What is the interaction of organisms with one another and with the environment called?

(a) Ecology

✓ (b) Ecosystem

(c) Habitat

(d) Food chain

2. (ii) What is the living of an organism in a particular environment called?

(a) Ecosystem

✓ (b) Habitat

(c) Temperature

(d) Ecosphere

2. (iii) What is the water of the area where the river meets the sea called?

(a) Fresh water

(b) Salty or sea water

✓ (c) Brackish water

(d) Binal water

2. (iv) What is a cold region covered with snow called?

(a) Marshy region

(b) Desert region

(c) Grassland

✓ (d) Tundra

2. (v) Where does a food chain always start?

(a) From the primary consumers

(b) From the second-degree or secondary consumers

✓ (c) From the producers

(d) From the decomposers

Water

1. (a) All the movement and circulation of water on the earth is called _____.

Answer: cycle.

1. (b) In water-cleaning plants, _____ is used to gather together the tiny particles of sand and pesticides.

Answer: A flocculant chemical

1. (c) To make water fit for drinking, it is boiled for _____.

Answer: 15 to 20 minutes, which kills the disease-causing germs.

1. (d) Sea weeds and sea plants are used to make Fertilizer. .

Question: 2. Match each item in column (A) with the correct statement in column (B): (A) 1. Potash alum, 2. Liquid chlorine, 3. Distillation.

Answer: 1. Potash alum — it cleans water by making the things present in the water stick together into large particles. 2. Liquid chlorine — it is used to destroy harmful microbes. 3. Distillation — it is used to separate water from dirt and physical and chemical things: the water is boiled into vapour and then cooled to obtain clean water.

Question: 3. (a) Why are Pakistan's fresh-water reserves running out? Give the reason.

Answer: Because fresh water is not preserved and protected for domestic, agricultural and industrial use, and water is being polluted by domestic, industrial and agricultural waste. Wastage of water and increasing use are also reducing the reserves.

Question: 3. (b) If water does not meet the standard of clean water, why does it cause diseases? Give the reason.

Answer: Because such water contains disease-causing germs, chemical substances and particles of pesticides, which cause hepatitis, cholera, typhoid, diarrhoea and other diseases.

Question: 4. Tell the ways by which we know that water is extremely necessary for animals.

Answer: Animals drink water every day; their food also contains water; without water animals cannot stay alive; plants too need water to make food (photosynthesis), and plants are the source of food for animals. So water is a basic need for life.

Question: 5. (a) Why is boiled water suitable for drinking and cooking? Explain.

Answer: Because boiling water for 15 to 20 minutes kills the disease-causing germs in it, so boiled water becomes safe for drinking and cooking.

Question: 5. (b) Why is salt separated from saline water? Explain.

Answer: Removing salt from saline water is called desalination. Such water is used for drinking, domestic work and recreation.

Question: 5. (c) Why is clean water necessary for making the environment pleasant? Explain.

Answer: Because clean water is necessary for the life of plants, animals and human beings, and the water cycle keeps the environment balanced. Dirty water spoils the environment and spreads diseases.

Question: 6. Describe five ways to reduce the wastage of water in daily life.

Answer: 1. Get leaking taps and pipes repaired at once. 2. Do not leave the tap running while brushing your teeth or washing your hands and face. 3. Use water carefully while bathing, and turn off the shower while applying soap. 4. Use a bucket instead of a hose pipe to wash a car. 5. Run

the washing machine only when it has a full load, and water the garden in the morning or evening so that the moisture stays in the soil for a long time.

Question: 7. Describe the processes (Processes) of cleaning water in water-cleaning plants (Plants).

Answer: 1. RO filter plant (Reverse Osmosis): water is passed through a semi-permeable membrane, which stops salts, chemical substances and germs. 2. Water treatment plant: first a flocculant chemical is mixed into the water to gather the tiny particles and settle them at the bottom, then the water is filtered through sand and stones, then chlorine is mixed in to kill disease-causing germs, and finally the water is sent to homes and factories.

Question: Project (a) Prepare a model of your own water-cleaning plant and label its parts.

Answer: Make a model showing a tank for dirty water, a flocculation section, a filter (sand and stones) section, a chlorination section and a clean-water tank. Put a label with the name on each part.

Question: Project (b) Prepare a list of the physical, chemical and biological things found in water.

Answer: Physical things: sand, clay, suspended particles. Chemical things: salts, pesticides (such as DDT), factory chemicals. Biological things: bacteria, viruses and other disease-causing germs.

Question: Project (c) Explain in detail how your prepared plant will work.

Answer: Dirty water will first enter the tank, where the flocculant chemical will gather the tiny particles and settle them at the bottom. Then the water will pass through the filter and become clean, then chlorine will kill the disease-causing germs, and finally the clean water will collect in the tank.

Question: Project (d) Make a poster advertising your water-cleaning plant and put it up in the school.

Answer: Make a poster showing the name of the plant, how it works and the benefits of clean water, and put it up in a prominent place in the school.

The Atom and Its Structure

Question: 1. (i) What kind of charge is there on the centre (nucleus) of an atom?

Answer: A positive charge, because the nucleus contains positively charged protons and neutrons without any charge.

Question: 1. (ii) Which particles keep revolving around the nucleus?

Answer: Electrons keep revolving around it. They revolve in different orbits or shells (K, L, M).

Question: 1. (iii) What is a neutral atom?

Answer: An atom in which the number of protons is equal to the number of electrons, meaning it has no overall charge, is called a neutral atom.

Question: 1. (iv) Show the structure of the atoms of the following elements: (a) Magnesium, (b) Silicon, (c) Sulphur, (d) Calcium, (e) Aluminium.

Answer: In each diagram show the nucleus (protons + neutrons) in the centre and shells around it. (a) Magnesium (12): 2 electrons in K, 8 in L, 2 in M. (b) Silicon (14): 2 in K, 8 in L, 4 in M. (c) Sulphur (16): 2 in K, 8 in L, 6 in M. (d) Calcium (20): 2 in K, 8 in L, 8 in M, 2 in N. (e) Aluminium (13): 2 in K, 8 in L, 3 in M.

Question: 1. (v) Tell the difference between cations (Cations) and anions (Anions).

Answer: When an atom gives away its electron it becomes positively charged and is called a cation (example: Na^{+}). When an atom gains an electron it becomes negatively charged and is called an anion (example: Cl^{-}).

Question: 2. Complete the following chart: for each element write its ion,

the number of protons and the number of electrons — Aluminium, Chlorine, Sodium, Potassium.

Answer: Aluminium: ion Al^{3+}, protons 13, electrons 10. Chlorine: ion Cl^{-}, protons 17, electrons 18. Sodium: ion Na^{+}, protons 11, electrons 10. Potassium: ion K^{+}, protons 19, electrons 18.

Question: 3. How many electrons are there in the valence shell of the following elements? (a) Nitrogen, (b) Helium, (c) Lithium.

Answer: (a) Nitrogen: 5 valence electrons (arrangement 2, 5). (b) Helium: 2 valence electrons (arrangement 2) — its only shell is complete. (c) Lithium: 1 valence electron (arrangement 2, 1).

Question: 4. Which element has only one electron in its outermost shell?

Answer: Elements such as hydrogen, lithium, sodium and potassium have only one electron in their outermost shell, so their valency is 1.

Question: 5. Write the number of carbon and hydrogen atoms in methane gas.

Answer: In methane (CH_4) there is 1 atom of carbon and 4 atoms of hydrogen.

Question: 6. Write the chemical formula of an ammonia molecule.

Answer: NH_3.

Question: 7. Complete the following chart: for each compound write its elements and the ratio of atoms — Water (H_2O), Aluminium oxide (Al_2O_3), Calcium oxide (CaO), Hydrochloric acid (HCl), Calcium fluoride (CaF_2).

Answer: Water: elements hydrogen and oxygen, ratio H:O = 2:1.

Aluminium oxide: elements aluminium and oxygen, ratio Al:O = 2:3.

Calcium oxide: elements calcium and oxygen, ratio Ca:O = 1:1.

Hydrochloric acid: elements hydrogen and chlorine, ratio H:Cl = 1:1.

Calcium fluoride: elements calcium and fluorine, ratio Ca:F = 1:2.

8. (i) Where is the whole mass/weight of an atom contained?

✓ (a) In the nucleus

(b) In the orbit

(c) In the neutron.

8. (ii) The mass of any atom is equal to _____.

(a) The mass of neutrons and electrons

(b) The mass of electrons and protons

✓ (c) The mass of neutrons and protons.

8. (iii) Atoms of the same element which have different mass numbers are called what?

(a) Anions

(b) Cations

✓ (c) Isotopes.

8. (iv) What does a chemical formula show?

(a) The number of atoms in a molecule

✓ (b) The number of atoms of each element

(c) The number of atoms in a compound.

8. (v) An atom has atomic number 16 and mass number 32; how many protons will it have?

✓ (a) 16

(b) 32

(c) 18.

Question: 9. Complete the following chart: for each row write the atomic number, mass number, number of protons, neutrons and electrons, the chemical symbol of the element, the name of the element and the isotope.

Answer: Row 1: atomic number 6, mass number 12, protons 6, neutrons 6, electrons 6, symbol C, element Carbon, isotope Carbon-12. Row 2: atomic number 12, mass number 24, protons 12, neutrons 12, electrons 12, symbol Mg, element Magnesium, isotope Magnesium-24. Row 3: atomic

number 17, mass number 35, protons 17, neutrons 18, electrons 17, symbol Cl, element Chlorine, isotope Chlorine-35. Row 4: atomic number 13, mass number 27, protons 13, neutrons 14, electrons 13, symbol Al, element Aluminium, isotope Aluminium-27.

Question: Project — Make models: make models of the atoms of Calcium, Magnesium, Sodium and Chlorine. Materials needed: clay or small balls of different colours and wire for the orbits.

Answer: Make each atom's model like this: a nucleus in the centre (balls of different colours for protons and neutrons) and wire orbits (shells) around it with balls for electrons on them. Calcium: 2 in K, 8 in L, 8 in M, 2 in N. Magnesium: 2 in K, 8 in L, 2 in M. Sodium: 2 in K, 8 in L, 1 in M. Chlorine: 2 in K, 8 in L, 7 in M.

Physical and Chemical Changes

Question: 1. (i) Tell the clear difference between physical and chemical changes, and give an example of each.

Answer: In a physical change only the shape, state, size or texture of the matter changes, but its chemical composition does not change, and the change is usually reversible — examples: ice turning into water, tearing paper into pieces. In a chemical change the chemical composition of the matter changes and a new thing is produced, whose properties are completely different from the original thing, and the change is usually irreversible — examples: paper burning, iron rusting, the process of photosynthesis.

Question: 1. (ii) Give examples of physical and chemical changes happening in your own home, school and locality.

Answer: Physical changes: ice melting, water boiling and turning into vapour, tearing paper into pieces, salt dissolving in water. Chemical changes: food cooking, a candle burning, iron rusting, milk going sour.

Question: 1. (iii) Tell the type of change produced in the following: (a) water turning into a solid thing (ice), (b) mixing things together and keeping them, (c) separating a mixture from one another.

Answer: (a) Physical change — only the state of the water changes, not its composition. (b) Physical change — mixing things together does not make any new thing. (c) Physical change — separating a mixture gives each thing back in its original state.

Question: 2. (i) Why are hydrocarbons (Hydrocarbons) used as fuel?

Answer: Because hydrocarbons give heat and light when they burn. Natural gas (methane), petrol, diesel and furnace oil are used as fuel in

homes, factories and vehicles.

Question: 2. (ii) What are fertilizers? What is their use? And why can they not be used in an improper way?

Answer: Fertilizers are things that give nutrients to the soil so that the crop grows well. Using them carelessly or in excess harms the environment, destroys the humus and fertility of the soil, and nitrate enters crops and fruits and harms people. So they should be used on the advice of an agriculture expert.

Question: 2. (iii) Describe the general properties of fertilizers.

Answer: Physical properties: particle size, density, granule hardness and moisture content. Chemical properties: the nutrients present in them — nitrogen, phosphorus, potassium, calcium, magnesium, sulphur, carbon, hydrogen and oxygen.

Question: 2. (iv) Describe the simple method of making plastic.

Answer: Mix borax powder in water in one cup; in another cup mix water and half a cup of white glue. Slowly mix the glue mixture into the borax water, add drops of colour and mix well. A soft, slimy substance (slime) will form, which is a simple plastic/polymer.

Question: 2. (v) Tell how vegetable oil turns into ghee.

Answer: By the process of hydrogenation: hydrogen is mixed into vegetable oil in the presence of a nickel catalyst, which turns the unsaturated oil into saturated fat (ghee/margarine).

3. (i) The melting of ice cream is a A physical type of change.

3. (ii) An irreversible change is called a chemical type of change.

3. (iii) The burning of a candle is a A chemical type of change.

3. (iv) Obtaining salt from sea water is a A physical type of change.

3. (v) A temporary change is called _____.

Answer: and can be reversed.

4. (i) When something burns, apart from heat, light and water vapours, what else is produced?

✓ (a) Carbon dioxide gas is produced

(b) Methane gas is produced

(c) Oxygen gas is produced.

4. (ii) What change takes place when wood rots or burns?

✓ (a) Chemical change

(b) Physical change

(c) Temporary change.

4. (iii) Which process is used to change vegetable oil into ghee?

(a) The process of evaporation

(b) The process of filtration

✓ (c) The process of hydrogenation.

4. (iv) What needs of plants are fulfilled by the use of fertilizer?

✓ (a) The deficiency of nutrients is fulfilled

(b) The needs of the breathing process are fulfilled

(c) The needs of reproduction are fulfilled.

Question: 5. Divide the following statements into reversible changes and irreversible changes: 1. Twisting rubber, 2. A candle burning, 3. Glass melting, 4. The process of photosynthesis, 5. Formation of glucose, 6. Formation of water by hydrogen and oxygen joining, 7. Ice turning into water, 8. Paper burning into ash, 9. Boiling an egg, 10. Mixture of salt and sand, 11. An electric bulb lighting up, 12. Dissolving sugar in water.

Answer: Reversible changes (physical): 1. Twisting rubber, 3. Glass

melting, 7. Ice turning into water, 10. Mixture of salt and sand, 11. An electric bulb lighting up, 12. Dissolving sugar in water. Irreversible changes (chemical): 2. A candle burning, 4. The process of photosynthesis, 5. Formation of glucose, 6. Formation of water by hydrogen and oxygen joining, 8. Paper burning into ash, 9. Boiling an egg.

Question: Project 1 — Plastic bottle in the sea: take a 1.5 litre plastic bottle, water, cooking oil and blue food colour, and observe.

Answer: Fill the bottle two-thirds with water, mix the blue colour into the water, then pour cooking oil on top, close the cap and shake the bottle.

Observation: the oil does not mix with the water; it keeps floating on top of the water and two separate layers are seen. This shows a situation like oil pollution in the sea, which is harmful for sea life.

Question: Project 2 — Dance of colours: pour milk in a dish, add drops of different colours to it, then touch the surface of the milk with a cotton bud dipped in soapy water, and observe.

Answer: On touching with soap, the colours start dancing and spreading around. Reason: the soap breaks the surface tension of the milk and reacts with the fat in the milk, which pushes the colours around. This is a combination of a physical and a chemical process.

Question: Project 2, question 1 — How were soap and the mixture used in this activity or project?

Answer: Soap was mixed in water, a cotton bud was dipped in it and used to touch the surface of the milk, so that the movement of the colours could be seen.

Question: Project 2, question 2 — What was the work of the soap and the water?

Answer: The water was the medium of the milk in which the colours floated, and the soap broke the surface tension of the milk and reacted

with the fat to set the colours moving.

Question: Project 2, question 3 — Why did the dance of colours start?

Answer: When the soapy bud touched the surface of the milk, the soap reacted with the fat and broke the surface tension, which pushed the colours around and made them dance.

Question: Project 2, question 4 — What chemical changes took place in this activity?

Answer: The soap reacting with the fat in the milk and the colours spreading in a new pattern show a combination of chemical and physical changes — the breaking of the surface tension is a physical process and the soap reacting with the fat is a chemical process.

Transfer of Heat

Question: 1. Match each statement in column (A) with the correct thing in column (B): (A) 1. The degree of hotness or coldness of a hot body, 2. A good conductor of heat, 3. Currents produced in the sea, 4. The process of heat transfer in a vacuum, 5. A poor radiator of heat.

Answer: 1. The degree of hotness or coldness of a hot body — (d) Temperature. 2. A good conductor of heat — (b) Copper (copper conducts heat well). 3. Currents produced in the sea — (e) Ocean currents. 4. The process of heat transfer in a vacuum — (c) Heat travelling by waves (Radiation). 5. A poor radiator of heat — (a) White colour (white colour absorbs little heat and radiates little heat).

2. (i) The cool air blowing from the land towards the sea in the evening is called _____.

Answer: sea.

2. (ii) A refrigerator works on the _____ method(s) of heat transfer.

Answer: evaporator fins take

2. (iii) An ocean current is a The convection method of heat transfer.

2. (iv) We get heat from the sun through the form of waves method.

2. (v) Insulators never Transfer/conduct heat.

Question: 3. (i) Why do we use a thermos in daily life? Give the reason.

Answer: Because a thermos keeps hot things hot and cold things cold for a long time. Its double-walled glass bottle, the vacuum in between, the silver coating and the stopper block all three methods of heat transfer —

conduction, convection and radiation.

Question: 3. (ii) Why does the process of conduction of heat take place only in solids? Give the reason.

Answer: Because in conduction the molecules vibrate in their fixed places and pass the heat to the neighbouring molecule. The molecules of solids are in fixed places, while the molecules of liquids and gases move about freely, so in them heat is transferred by convection.

Question: 3. (iii) Why does the heat of the sun not reach the earth by the methods of conduction and currents?

Answer: Because between the sun and the earth there is a vacuum, where no matter/medium is present. Conduction and convection need a material medium, while radiation needs no medium, so the sun's heat reaches us only by radiation.

Question: 4. Describe five examples of the use of radiation in your daily life.

Answer: 1. The sun's heat reaches the earth by waves. 2. Standing near a fire or a heater, heat is felt without touching it. 3. In light/white clothes less heat is felt in summer, because a light colour reflects heat back. 4. Blackening the bottom of a kettle makes it absorb more heat. 5. Petrol tankers are painted white or silver so that they reflect the sun's heat back and keep the petrol from getting hot.

Question: Project — Do all colours absorb equal heat? Wrap four boxes with papers of different colours (white, red or yellow, yellow, black), put an equal ice cube in each, and keep them in sunlight. Observe which colour box's ice melts first, second, third and fourth.

Answer: Result: the ice in the black box melts first, because black colour absorbs the most heat. After that the ice in the red/yellow box melts. The ice in the white box melts last, because white colour absorbs little heat

and reflects most of the heat back. So white colour is best for protecting ice from melting quickly.

Spreading of Light

1. (i) When light passes through a reflecting prism, it strikes at an angle which is _____ than the critical angle.

Answer: greater

1. (ii) The process of light splitting up into its different colours is called _____.

Answer: seven colours.

1. (iii) The critical angle of water is _____48.8 degrees._____.

1. (iv) In binoculars, _____ prisms are used to see distant objects.

Answer: Reflecting prisms. They work on the principle of total internal reflection and turn the rays of light through up

1. (v) If a body absorbs all the colours of light, it will appear _____ in colour.

Answer: appears black.

Question: 2. Write any five uses of the different colours of light other than the uses given in the book.

Answer: This question asks for five new uses besides those given in the book, and those five extra uses are not given in the book. The uses given in the book are: at home, making rooms pleasant and choosing furniture colours; at school, colours for classrooms and boards; on roads, signals, drivers' instructions and warnings; colours on buildings; colours on alarms and ambulance/fire-brigade vehicles; and coloured indicators on devices (such as the Wi-Fi or internet signal). No use beyond these is part of the book.

Question: 3. Draw a labelled diagram of the refraction of light going from

glass into water.

Answer: This is a diagram-drawing task; the diagram must show: a glass semi-circle (denser medium) with water above it (rarer medium), the boundary between the two, a straight normal line at the boundary, the incident ray coming through the glass, the refracted ray entering the water bent away from the normal (because light is going from a denser medium into a rarer medium), and the labels: glass, water, normal, incident ray, refracted ray, angle of incidence, angle of refraction.

Question: 4. (i) State the difference between real depth and apparent depth.

Answer: Real depth is the actual depth at which an object lies in water. Apparent depth is the depth at which the object appears to lie. Because of the refraction of light, an object under water appears higher up and shallower than its real position, so the apparent depth is less than the real depth.

Question: 4. (ii) State the difference between primary colours and secondary colours.

Answer: There are three primary colours: red, green and blue. Secondary colours are the colours obtained by mixing two primary colours together in the correct proportions.

Question: 5. (a) State the laws of refraction of light.

Answer: There are two laws of refraction: (1) The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane. (2) The ratio between the speed of light in a vacuum and the speed of light in a medium is always constant; this ratio is called the refractive index of that medium. These laws are also called Snell's laws.

Question: 5. (b) Explain the refractive index with two examples.

Answer: Refractive index (n) = speed of light in a vacuum ÷ speed of light

in the medium. Example: the refractive index of water is 1.33, which means light travels 1.33 times more slowly in water than in a vacuum. The refractive index of diamond is 2.42, which means light travels 2.42 times more slowly in diamond than in a vacuum.

Question: 6. (i) Explain the condition of a mirage.

Answer: A mirage is an illusion seen in a desert or on a hot road. The layers of hot air have different densities, so light passing through them bends and undergoes total internal reflection, which makes the image of a distant object look as if water were standing there.

Question: 6. (ii) Explain the condition of the fish eye view.

Answer: A fish sitting inside the water sees the world above through total internal reflection. The sun, the moon, birds and other animals appear in a circle at the surface of the water, but because of total internal reflection they look different from their real shape and position.

Question: Project (a): Prepare a list showing real events of refraction of light, total internal reflection and dispersion of light.

Answer: This is a practical activity with no single fixed answer. The list can use these examples from the book: refraction of light — a pencil looking bent in water; total internal reflection — mirage, the fish eye view, binoculars and periscope; dispersion of light — the rainbow.

Question: Project (b): Explain the conditions in which the processes of refraction of light, total internal reflection and dispersion appear.

Answer: This is a practical activity with no single fixed answer. The explanation should be based on the book's conditions: refraction happens when light goes from one medium into another and its speed changes; total internal reflection happens when light goes from a denser medium into a rarer medium and the angle of incidence becomes greater than the critical angle; dispersion happens when white light passes through a prism

and splits into seven colours.

Question: Project (c): Describe one event in which an action of light appears.

Answer: This is a practical activity with no single fixed answer. Example: the appearance of a rainbow in the sky after rain — this is the process of dispersion of light, in which the white light of the sun passes through the drops of water present in the air and splits into seven colours.

Question: Project (d): Explain clearly what effect it would have on human life if the processes of refraction of light, total internal reflection and dispersion did not occur naturally in our surroundings.

Answer: This is a practical activity with no single fixed answer. Points to think about: if there were no refraction, estimates of the depth and position of things in water would be wrong; if there were no total internal reflection, instruments like binoculars and periscopes would not work and sights like mirages would not be seen; if there were no dispersion, no rainbow would form and no spectrum of colours would be obtained from a prism.

Sound Waves

1. (i) What is the type of sound and light waves?

(a) Only transverse waves

(b) Only longitudinal waves

✓ (c) Both transverse and longitudinal waves

1. (ii) By which symbol is the wave length shown?

(a) V

✓ (b) λ

(c) δ

1. (iii) Sounds of which frequency range can the human ear hear?

(a) 16-20,000 Hz

✓ (b) 20-20,000 Hz

(c) 20-3,000 Hz

1. (iv) Through which thing does sound pass fastest of all?

(a) Air

(b) Water

✓ (c) Steel

1. (v) Decibel is the unit of measurement of which thing?

(a) Pitch

✓ (b) Amplitude

(c) Wave length

Question: 2. On what do the quality and the intensity of sound depend?

Answer: The quality of sound depends on the shape of the source producing the sound. The intensity of sound depends on the amplitude of the sound waves — the greater the amplitude, the louder the sound.

Question: 3. Explain the importance of sound for humans through five

means.

Answer: Five examples from the daily uses given in the book: (1) Communication — talking and messaging through mobile phones and telephones; (2) Safety and warning — the doorbell, the smoke-warning device, sirens and alarms; (3) Entertainment — radio and the music system; (4) Traffic and announcements — the horn, the azan or call, traffic sounds and announcements; (5) Helping devices — the car's parking sensor, which warns of obstacles through sound.

Question: 4. The minimum and maximum limits of sound frequency are given below. From these limits, state the hearing range of the following living things: young woman, old man, bat, cat, whale, dog, dolphin, elephant. Ranges: 20-15,000 Hz; 2,000-110,000 Hz; 20-120,000 Hz; 45-64,000 Hz.

Answer: According to the book's table, three ranges match these animals: 2,000-110,000 Hz — bat; 20-120,000 Hz — dolphin; 45-64,000 Hz — cat. The 20-15,000 Hz range is not given for any animal in the book's table (the book gives the human range as 20-20,000 Hz), so it cannot be matched to a particular animal on the book's basis. The remaining names (young woman, old man, whale, dog, elephant) do not match these four ranges in the book — the book gives whale as 10-31,000 Hz, dog as 67-45,000 Hz and elephant as 16-20,000 Hz.

Question: Project: Prepare a xylophone from five bottles (bottle V with 400 ml, bottle IV with 350 ml, bottle III with 250 ml, bottle II with 200 ml and bottle I with 100 ml of water) and state what relation the shape of the xylophone has with the musical sound produced by it.

Answer: This is a practical activity with no single fixed answer. Method: take five bottles of different sizes, mark them I to V, fill each bottle with the given amount of water, then arrange them in this order: IV, II, V, I, III,

and strike each bottle with a spoon and listen to the sound. Result: each bottle gives a different sound, because according to the book the quality of sound depends on the shape of the source producing the sound, and the pitch depends on the frequency of the sound — changing the amount of water in the bottle and the shape of the bottle changes the pitch and the quality of the sound.

Electric Circuits and Electric Current

1. (a) An electric circuit in which there is only one single path for the current to flow through is called _____.

Answer: current.

1. (b) Electric current is measured in amperes (A). units.

1. (c) The unit for measuring the potential difference is _____.

Answer: measured in volts (V).

1. (d) In homes and in the kitchen, the _____ type of circuit is fitted.

Answer: current, and each device can be switched on or off separately.

1. (e) The _____ unit is used for measuring resistance.

Answer: Ohm (Ω). Resistance

Question: 2. (a) What is electric current?

Answer: Electric current is the movement of free electrons. When electrons flow in one direction through a conductor, a flow of electric charges is produced, which is called electric current.

Question: 2. (b) What is it that flows as electric current in an electric circuit?

Answer: Electric charges (electrons) flow. The electrons freed from the outer shell of a conductor flow through the circuit, and this flow itself is the current.

Question: 2. (c) What is the difference between a series circuit and a parallel circuit?

Answer: In a series circuit there is only one path for the current, and the same amount of current flows through the whole circuit; if one bulb fuses, the whole circuit stops. In a parallel circuit there is more than one path

(branch) for the current; if one branch is switched off, current keeps flowing in the other branches.

Question: 2. (d) Tell what type of electric circuit is fitted in your classroom. Explain it too.

Answer: A parallel circuit. In the classroom every bulb and fan can be switched on or off with its own separate switch, and if one bulb fuses the other bulbs keep glowing — that is the sign of a parallel circuit, in which there is more than one path for the current.

Question: 3. What accidents can happen because of electricity?

Answer: Three main accidents are: (1) Electric shock and electrocution — when current passes through the body a shock is felt, and a person can be injured or can even die. (2) Short circuit — when too much current passes through wires, the wires heat up and touch each other, which produces sparks and can start a fire. (3) Fire — hot wires and sparks can set nearby things on fire.

Question: 4. Tell the ways of using electricity in a safe manner.

Answer: (1) Do not put any metal thing into electric sockets. (2) Never tamper with an electric wire, and never pull it out of the power-house wire. (3) Do not touch running electric things with wet hands. (4) Do not plug more things than the limit into sockets. (5) Do not touch a person who has received an electric shock with your hands, and do not try to rescue them by hand; move them away from the wire with the help of a non-metal thing (such as wood or rope). Besides this, use safety devices: fuse, MCB, earth wire, three-pin plug and ELCB.

Exploring Space

1. (a) The scientists' theory which explains the _____ of the universe is called the Big Bang.

Answer: tells that

1. (b) Stars are actually big round balls of gas, which mainly contain _____ and _____.

Answer: hydrogen, helium gas.

1. (c) The temperature of blue-coloured stars is _____ than the temperature of red, yellow or orange-coloured stars.

Answer: are

1. (d) A galaxy is a huge combined collection of _____, _____ and _____.

Answer: stars,, gas, fine particles (dust), held together by the force of gravity.

1. (e) The lenticular galaxy is an in-between type of _____ and _____ galaxies in shape.

Answer: elliptical, galaxy.

1. (f) A black hole is actually a place in space where _____ collapses onto itself and even _____ cannot escape from it.

Answer: hole, matter collects so much that its gravity becomes so strong that, light

1. (g) The telescope is the instrument which can collect the _____ coming from faraway things.

Answer: light

✗ False: 2. (a) This universe came into existence about 10 billion years ago. (True / False)

✗ False: 2. (b) CMB is used for Cosmic Matter Blue Radiation. (True / False)

✓ True: 2. (c) NASA sent a satellite named COBE into space in 1992. (True / False)

✓ True: 2. (d) The temperature of the Sun is about 6000 degrees Celsius. (True / False)

✗ False: 2. (e) The star named Arcturus is the hottest of all stars. (True / False)

✓ True: 2. (f) The name of the galaxy in which the Earth is present is the Milky Way. (True / False)

✗ False: 2. (g) The scientist Einstein presented the idea of the classification of galaxies. (True / False)

✗ False: 2. (h) Our Sun is located about 15,000 light years away from the centre of the galaxy. (True / False)

✗ False: 2. (i) The last stage of the life of a Sun-like star is a black hole. (True / False)

✓ True: 2. (j) Constellations in the sky only reflect an imaginary body. (True / False)

Question: 3. (a) What type of star ends up in the form of a black hole at the end of its life?

Answer: A very massive big star. When a very big star ends with a supernova explosion, its remaining matter collapses onto itself and becomes a black hole.

Question: 3. (b) What is a light year?

Answer: A light year is a unit for measuring distance. The distance that light covers in one year is called one light year.

Question: 3. (c) Give the definition of an asterism.

Answer: The definition of an asterism is not given in the text of this

chapter, so its answer cannot be given from the book. The book only tells about a constellation, which is a group of stars in the sky that, seen from the Earth, shows the pattern or shape of something.

Question: 3. (d) On what conditions does the brightness of any star depend?

Answer: The brightness of a star depends on four things: (1) The star's temperature — a hotter star shines more. (2) The star's size — a bigger star shines more. (3) The star's distance from the Earth — a nearer star looks brighter. (4) The matter in between — the dust and gas present between the star and the Earth reduce the light and lessen the brightness.

Question: 3. (e) How many millions are there in one billion?

Answer: There are 1000 million in one billion. One million has 6 zeros after 1 (1,000,000), and one billion has 9 zeros after 1 (1,000,000,000).

Question: 3. (f) What is Cosmic Microwave Background Radiation?

Answer: The CMB is the leftover echo of the Big Bang. It is very weak microwave light/radiation which came out of the hot universe at the beginning of the universe and is still spread in every direction in space. It is the biggest proof in support of the Big Bang theory.

Question: 4. (a) Describe the coming into existence of the universe and the Big Bang theory.

Answer: The Big Bang theory tells that the universe began about 13.7 billion years ago from a single point. That point was extremely hot and dense, and it contained all the matter of the universe. A huge explosion took place in that point, from which space, time and matter came into existence, and the universe began to expand. The universe is still expanding, and it was hot at the beginning and then kept cooling down. There are two big proofs of this theory: (1) the expansion of the universe every day, (2) the Cosmic Microwave Background Radiation (CMB), which

is the leftover echo of the Big Bang.

Question: 4. (b) Describe the different stages of the life of low-mass stars.

Answer: Low-mass (Sun-like) stars pass through these stages: (1) Nebula — a cloud of gas and dust, containing 97 percent hydrogen and 3 percent helium. (2) Proto-star phase — under the force of gravity the cloud shrinks and becomes a hot ball. (3) Main sequence phase — the longest and stable phase of the star's life, in which hydrogen burns and becomes helium. (4) Red giant phase — when the hydrogen finishes, the star swells and becomes a very big red ball. (5) Planetary nebula phase — the star throws off its outer matter and takes the shape of a planetary nebula. (6) White dwarf — in the end the centre of the star is left as a small, dense and white body.

Question: 4. (c) Describe the different types of galaxies.

Answer: According to the Hubble sequence there are four main types of galaxies: (1) Spiral — these have several spiral-like structures or arms and a bright centre in the middle; our Milky Way is also spiral. (2) Elliptical — these are oval (egg-like) in shape and mostly contain old stars. (3) Lenticular — lens-shaped, which have a disc but no spiral arms. (4) Irregular — which have no fixed shape. Besides these, the book also mentions starburst galaxies, merging galaxies, active galaxies and radio galaxies.

Question: 4. (d) How do we measure the distance of nearby stars from the Earth?

Answer: The distance of nearby stars is measured by the parallax method. A nearby star is observed from two opposite places of the Earth's orbit, with a gap of six months. On both occasions that star appears slightly shifted against the background of the faraway stars; this apparent change is called parallax. From the measurement of the parallax, the distance of

the star from the Earth is found out, which is stated in light years.